

In the Name of God

Computational Fluid Dynamics II

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Chapter 2

The Mathematical Formulation of Euler Equations

فصل ۲

فرمولاسیون ریاضی معادلات اوایلر

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (خاصیت همگنی)

$F(x, y)$ is homogenous function of degree n if $F(\alpha x, \alpha y) = \alpha^n F(x, y)$

Euler theorem: $F(\alpha x, \alpha y) = \alpha^n F(x, y) \quad \rightarrow \quad x \frac{\partial F}{\partial x} + y \frac{\partial F}{\partial y} = n F(x, y)$

For system of eqs.: $\vec{F} = \vec{F}(\vec{x}) \quad \& \quad \begin{cases} \vec{x} = (x_1, x_2, x_3, \dots, x_n) = \text{solution vector} \\ \vec{F} = (F_1, F_2, F_3, \dots, F_n) = \text{flux vector} \end{cases}$

$$\begin{cases} x_1 \frac{\partial F_1}{\partial x_1} + x_2 \frac{\partial F_1}{\partial x_2} + \dots + x_n \frac{\partial F_1}{\partial x_n} = n F_1 \\ x_1 \frac{\partial F_2}{\partial x_1} + x_2 \frac{\partial F_2}{\partial x_2} + \dots + x_n \frac{\partial F_2}{\partial x_n} = n F_2 \\ \vdots \\ x_1 \frac{\partial F_n}{\partial x_1} + x_2 \frac{\partial F_n}{\partial x_2} + \dots + x_n \frac{\partial F_n}{\partial x_n} = n F_n \end{cases} \quad \rightarrow \quad A\vec{x} = n\vec{F} \quad \& \quad A = \frac{\partial \vec{F}}{\partial \vec{x}} = \text{Jacobian matrix}$$

فرمولاسیون ریاضی معادلات اوایلر

- معادلات اوایلر (فرم دوبعدی غیردائم تراکم پذیر)

2D unsteady compressible Euler eqs.:

$$\underbrace{\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y}}_{\text{conservative form}} = 0 \quad \& \quad \underbrace{Q = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E_t \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{pmatrix}}_{\text{conservative variables}} \quad \& \quad \underbrace{F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{pmatrix}}_{\text{flux vector}} \quad \& \quad \underbrace{G = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (E_t + p)v \end{pmatrix} = \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \end{pmatrix}}_{\text{flux vector}}$$

$$F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (E_t + p)u \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{pmatrix} = \begin{pmatrix} Q_2 \\ \frac{Q_2^2}{Q_1} + (\gamma - 1) \left\{ Q_4 - \frac{1}{2} \frac{Q_2^2 + Q_3^2}{Q_1} \right\} \\ \frac{Q_2 Q_3}{Q_1} \\ \left(Q_4 + (\gamma - 1) \left\{ Q_4 - \frac{1}{2} \frac{Q_2^2 + Q_3^2}{Q_1} \right\} \right) \frac{Q_2}{Q_1} \end{pmatrix} \quad \& \quad F(\alpha Q) = \alpha^n F(Q) \quad \& \quad n = 1$$

$$E_t = \rho \left(e + \frac{1}{2} (u^2 + v^2) \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho (u^2 + v^2) \quad \& \quad p = (\gamma - 1) \left\{ E_t - \frac{1}{2} \rho (u^2 + v^2) \right\} = (\gamma - 1) \left\{ Q_4 - \frac{1}{2} \frac{Q_2^2 + Q_3^2}{Q_1} \right\}$$

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (فرم دوبعدی غیردائم تراکم پذیر)

$$\begin{cases} AQ = nF \\ BQ = nG \end{cases} \quad \& \quad n=1 \quad \rightarrow \quad \begin{cases} AQ = F & \& \quad A = \frac{\partial F}{\partial Q} \\ BQ = G & \& \quad B = \frac{\partial G}{\partial Q} \end{cases}$$

$$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial Q} \frac{\partial Q}{\partial x} = A \frac{\partial Q}{\partial x} \quad \& \quad \frac{\partial F}{\partial x} = \frac{\partial(AQ)}{\partial x} = A \frac{\partial Q}{\partial x} \quad \rightarrow \quad \frac{\partial A}{\partial x} Q = 0$$

$$\frac{\partial G}{\partial y} = \frac{\partial G}{\partial Q} \frac{\partial Q}{\partial y} = B \frac{\partial Q}{\partial y} \quad \& \quad \frac{\partial G}{\partial y} = \frac{\partial(BQ)}{\partial y} = B \frac{\partial Q}{\partial y} \quad \rightarrow \quad \frac{\partial B}{\partial y} Q = 0$$

$$\underbrace{\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y}}_{\text{conservative form}} = 0 \quad \rightarrow \quad \underbrace{\frac{\partial Q}{\partial t} + A \frac{\partial Q}{\partial x} + B \frac{\partial Q}{\partial y}}_{\text{nonconservative form}} = 0$$

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (فرم یک بعدی غیردائم تراکم پذیر)

1D unsteady compressible Euler eqs.:

$$\underbrace{\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x}}_{\text{conservative form}} = 0 \quad \& \quad \underbrace{Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \\ Q_3 \end{pmatrix}}_{\text{conservative variables}} \quad \& \quad \underbrace{F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix}}_{\text{flux vector}}$$

$$F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} Q_2 \\ \frac{Q_2^2}{Q_1} + (\gamma - 1) \left\{ Q_3 - \frac{1}{2} \frac{Q_2^2}{Q_1} \right\} \\ \left(Q_3 + (\gamma - 1) \left\{ Q_3 - \frac{1}{2} \frac{Q_2^2}{Q_1} \right\} \right) \frac{Q_2}{Q_1} \end{pmatrix}$$

$$E_t = \rho \left(e + \frac{1}{2} u^2 \right) = \frac{p}{\gamma - 1} + \frac{1}{2} \rho u^2 \quad \& \quad p = (\gamma - 1) \left\{ E_t - \frac{1}{2} \rho u^2 \right\} = (\gamma - 1) \left\{ Q_3 - \frac{1}{2} \frac{Q_2^2}{Q_1} \right\} \quad \& \quad a^2 = \frac{\gamma p}{\rho} = \gamma RT$$

فرمولاسیون ریاضی معادلات اوایلر

- معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- ماتریس ژاکوبین متغیرهای بقایی
 $A = \frac{\partial F}{\partial Q}$ = Jacobian matrix of conservative variables

$$A_{11} = \frac{\partial F_1}{\partial Q_1} = \frac{\partial Q_2}{\partial Q_1} = 0, \quad A_{12} = \frac{\partial F_1}{\partial Q_2} = \frac{\partial Q_2}{\partial Q_2} = 1, \quad A_{13} = \frac{\partial F_1}{\partial Q_3} = \frac{\partial Q_2}{\partial Q_3} = 0$$

$$A_{21} = \frac{\partial F_2}{\partial Q_1} = \frac{\partial \left(\frac{Q_2^2}{Q_1} + (\gamma - 1) \left\{ Q_3 - \frac{1}{2} \frac{Q_2^2}{Q_1} \right\} \right)}{\partial Q_1} = -\frac{Q_2^2}{Q_1^2} + (\gamma - 1) \left(\frac{1}{2} \frac{Q_2^2}{Q_1^2} \right) = (\gamma - 3) \frac{u^2}{2}$$

$$A_{22} = \frac{\partial F_2}{\partial Q_2} = (3 - \gamma)u, \quad A_{23} = \frac{\partial F_2}{\partial Q_3} = \gamma - 1$$

$$A_{31} = \frac{\partial F_3}{\partial Q_1} = (\gamma - 1)u^3 - \gamma \frac{E_t u}{\rho}, \quad A_{32} = \frac{\partial F_3}{\partial Q_2} = \gamma \frac{E_t}{\rho} - \frac{3}{2}(\gamma - 1)u^2, \quad A_{33} = \frac{\partial F_3}{\partial Q_3} = \gamma u$$

فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- فرم بقایی و غیربقایی

$$\underbrace{\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x}}_{\text{conservative form}} = \frac{\partial Q}{\partial t} + \frac{\partial F}{\partial Q} \frac{\partial Q}{\partial x} = \underbrace{\frac{\partial Q}{\partial t} + A \frac{\partial Q}{\partial x}}_{\text{nonconservative form}} = 0$$

$$Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix}, \quad A = \frac{\partial F}{\partial Q} = \begin{pmatrix} 0 & 1 & 0 \\ (\gamma-3)\frac{u^2}{2} & (3-\gamma)u & \gamma-1 \\ (\gamma-1)u^3 - \gamma\frac{E_t u}{\rho} & \gamma\frac{E_t}{\rho} - \frac{3}{2}(\gamma-1)u^2 & \gamma u \end{pmatrix}$$

- فرم متغیرهای اولیه

$$\begin{cases} \rho_t + u\rho_x + \rho u_x = 0 \\ u_t + uu_x + \frac{1}{\rho} p_x = 0 \\ p_t + up_x + \rho a^2 u_x = 0 \end{cases} \rightarrow \underbrace{\frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x}}_{\text{primitive form}} = 0 \quad \& \quad \underbrace{\tilde{Q} = \begin{pmatrix} \rho \\ u \\ p \end{pmatrix}}_{\text{primitive variables}} \quad \& \quad \underbrace{\tilde{A} = \begin{pmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & \rho a^2 & u \end{pmatrix}}_{\text{Jacobain matrix of primitive variables}}$$

فرمولاسیون ریاضی معادلات اوایلر

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = \frac{\partial Q}{\partial t} + \frac{\partial F}{\partial Q} \frac{\partial Q}{\partial x} = \frac{\partial Q}{\partial t} + A \frac{\partial Q}{\partial x} = 0$$

$$A \quad \tilde{A}$$

$$\frac{\partial Q}{\partial \tilde{Q}} \frac{\partial \tilde{Q}}{\partial t} + A \frac{\partial Q}{\partial \tilde{Q}} \frac{\partial \tilde{Q}}{\partial x} = 0 \quad \& \quad M = \frac{\partial Q}{\partial \tilde{Q}} \rightarrow$$

$$M \frac{\partial \tilde{Q}}{\partial t} + AM \frac{\partial \tilde{Q}}{\partial x} = 0 \rightarrow$$

$$\frac{\partial \tilde{Q}}{\partial t} + M^{-1} AM \frac{\partial \tilde{Q}}{\partial x} = \frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = 0 \quad \& \quad \tilde{A} = M^{-1} AM$$

$$M = \frac{\partial Q}{\partial \tilde{Q}} = \begin{pmatrix} 1 & 0 & 0 \\ u & \rho & 0 \\ \frac{u^2}{2} & \rho u & \frac{1}{\gamma-1} \end{pmatrix} \quad \& \quad M^{-1} = \frac{\partial \tilde{Q}}{\partial Q} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{u}{\rho} & \frac{1}{\rho} & 0 \\ (\gamma-1)\frac{u^2}{2} & -(\gamma-1)u & \gamma-1 \end{pmatrix}$$

فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر با چشمه)

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = H$$

- تعیین فرم مشخصه‌ای سیستم معادلات

$$\frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = \tilde{H} \quad \& \quad \tilde{A} = M^{-1} A M \quad \& \quad \tilde{H} = M^{-1} H$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{Q}_1 \\ \tilde{Q}_2 \\ \tilde{Q}_3 \end{pmatrix}_t + \begin{pmatrix} \tilde{A}_{11} & \tilde{A}_{12} & \tilde{A}_{13} \\ \tilde{A}_{21} & \tilde{A}_{22} & \tilde{A}_{23} \\ \tilde{A}_{31} & \tilde{A}_{32} & \tilde{A}_{33} \end{pmatrix} \begin{pmatrix} \tilde{Q}_1 \\ \tilde{Q}_2 \\ \tilde{Q}_3 \end{pmatrix}_x = \begin{pmatrix} \tilde{H}_1 \\ \tilde{H}_2 \\ \tilde{H}_3 \end{pmatrix}$$

- ضرب سیستم معادلات در بردار

$$L^T = (\tilde{l}_1, \tilde{l}_2, \tilde{l}_3)$$

$$(\tilde{l}_1, \tilde{l}_2, \tilde{l}_3) \frac{\partial \tilde{Q}}{\partial t} + (\tilde{l}_1, \tilde{l}_2, \tilde{l}_3) \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = (\tilde{l}_1, \tilde{l}_2, \tilde{l}_3) \tilde{H}$$

$$\tilde{l}_1 \frac{\partial \tilde{Q}_1}{\partial t} + \tilde{l}_2 \frac{\partial \tilde{Q}_2}{\partial t} + \tilde{l}_3 \frac{\partial \tilde{Q}_3}{\partial t} +$$

$$\underbrace{(\tilde{l}_1 \tilde{A}_{11} + \tilde{l}_2 \tilde{A}_{21} + \tilde{l}_3 \tilde{A}_{31})}_{\tilde{l}_1 \lambda_1} \frac{\partial \tilde{Q}_1}{\partial x} + \underbrace{(\tilde{l}_1 \tilde{A}_{12} + \tilde{l}_2 \tilde{A}_{22} + \tilde{l}_3 \tilde{A}_{32})}_{\tilde{l}_2 \lambda_1} \frac{\partial \tilde{Q}_2}{\partial x} + \underbrace{(\tilde{l}_1 \tilde{A}_{13} + \tilde{l}_2 \tilde{A}_{23} + \tilde{l}_3 \tilde{A}_{33})}_{\tilde{l}_3 \lambda_1} \frac{\partial \tilde{Q}_3}{\partial x} = \tilde{l}_1 \tilde{H}_1 + \tilde{l}_2 \tilde{H}_2 + \tilde{l}_3 \tilde{H}_3$$

فرمولاسیون ریاضی معادلات اوایلر

- معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر با چشمه)

- تعیین فرم مشخصه‌ای سیستم معادلات

$$\tilde{l}_1 \left(\frac{\partial \tilde{Q}_1}{\partial t} + \lambda_1 \frac{\partial \tilde{Q}_1}{\partial x} \right) + \tilde{l}_2 \left(\frac{\partial \tilde{Q}_2}{\partial t} + \lambda_1 \frac{\partial \tilde{Q}_2}{\partial x} \right) + \tilde{l}_3 \left(\frac{\partial \tilde{Q}_3}{\partial t} + \lambda_1 \frac{\partial \tilde{Q}_3}{\partial x} \right) = \tilde{l}_1 \tilde{H}_1 + \tilde{l}_2 \tilde{H}_2 + \tilde{l}_3 \tilde{H}_3$$

$$\text{On } C: \frac{dx}{dt} = \lambda_1$$

- معادلات سازگاری

$$\tilde{l}_1 \frac{d\tilde{Q}_1}{dt} + \tilde{l}_2 \frac{d\tilde{Q}_2}{dt} + \tilde{l}_3 \frac{d\tilde{Q}_3}{dt} = \tilde{l}_1 \tilde{H}_1 + \tilde{l}_2 \tilde{H}_2 + \tilde{l}_3 \tilde{H}_3 \quad \text{compatibility eqs}$$

$$\tilde{l}_j \frac{d\tilde{Q}_j}{dt} = \tilde{l}_j \tilde{H}_j \quad \text{on} \quad C_1: \frac{dx}{dt} = \lambda_1$$

فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر با چشمه)

$$\begin{cases} \tilde{l}_1 \tilde{A}_{11} + \tilde{l}_2 \tilde{A}_{21} + \tilde{l}_3 \tilde{A}_{31} = \tilde{l}_1 \lambda_1 \\ \tilde{l}_1 \tilde{A}_{12} + \tilde{l}_2 \tilde{A}_{22} + \tilde{l}_3 \tilde{A}_{32} = \tilde{l}_2 \lambda_1 \\ \tilde{l}_1 \tilde{A}_{13} + \tilde{l}_2 \tilde{A}_{23} + \tilde{l}_3 \tilde{A}_{33} = \tilde{l}_3 \lambda_1 \end{cases} \rightarrow \tilde{l}_i \tilde{A}_{ij} = \tilde{l}_j \lambda_1$$

$$L^{(i)} = \begin{pmatrix} \tilde{l}_1 \\ \tilde{l}_2 \\ \tilde{l}_3 \end{pmatrix}^{(i)} \rightarrow (L^i)^T [\tilde{A}] = \lambda (L^i)^T \quad \& \quad (L^i)^T [\tilde{A} - \lambda I] = 0$$

- فرم قطری ماتریس

$$L^T = [L]^T = [L^{(1)}, L^{(2)}, L^{(3)}]^T = \begin{pmatrix} \tilde{l}_1^{(1)} & \tilde{l}_2^{(1)} & \tilde{l}_3^{(1)} \\ \tilde{l}_1^{(2)} & \tilde{l}_2^{(2)} & \tilde{l}_3^{(2)} \\ \tilde{l}_1^{(3)} & \tilde{l}_2^{(3)} & \tilde{l}_3^{(3)} \end{pmatrix} \quad \& \quad L^T \tilde{A} = D L^T$$

$$L^T \tilde{A} = D L^T \rightarrow L^T \tilde{A} (L^T)^{-1} = D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر با چشمه)

$$\frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = \tilde{H} \quad \& \quad \tilde{H} = M^{-1} H$$

$$L^T \frac{\partial \tilde{Q}}{\partial t} + L^T \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = L^T \tilde{H}$$

$$L^T \frac{\partial \tilde{Q}}{\partial t} + \underbrace{L^T \tilde{A} (L^T)^{-1}}_D L^T \frac{\partial \tilde{Q}}{\partial x} = L^T \tilde{H}$$

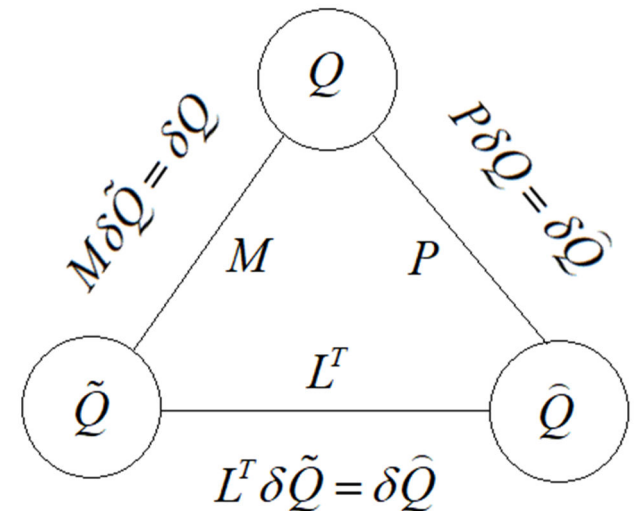
$$L^T \frac{\partial \tilde{Q}}{\partial t} + D L^T \frac{\partial \tilde{Q}}{\partial x} = L^T \tilde{H} \quad \text{compatibility eqs. (ODE)}$$

$$\frac{\partial \hat{Q}}{\partial t} + D \frac{\partial \hat{Q}}{\partial x} = L^T \tilde{H} \quad \& \quad \delta \hat{Q} = L^T \delta \tilde{Q} \quad \& \quad \delta \equiv \partial_x \text{ or } \partial_t$$

$$L^T \frac{\partial \tilde{Q}}{\partial Q} \frac{\partial Q}{\partial t} + D L^T \frac{\partial \tilde{Q}}{\partial Q} \frac{\partial Q}{\partial x} = L^T \tilde{H}$$

$$L^T M^{-1} \frac{\partial Q}{\partial t} + D L^T M^{-1} \frac{\partial Q}{\partial x} = L^T M^{-1} H$$

$$P \frac{\partial Q}{\partial t} + D P \frac{\partial Q}{\partial x} = P H \quad \& \quad \delta \hat{Q} = P \delta Q \quad \& \quad P = L^T M^{-1}$$



فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- تعیین فرم مشخصه‌ای سیستم معادلات

$$\begin{cases} \rho_t + u\rho_x + \rho u_x = 0 \\ u_t + uu_x + \frac{1}{\rho}p_x = 0 \\ p_t + up_x + \rho a^2 u_x = 0 \end{cases} \rightarrow \underbrace{\frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x}}_{\text{primitive form}} = 0 \quad \& \quad \underbrace{\tilde{Q} = \begin{pmatrix} \rho \\ u \\ p \end{pmatrix}}_{\text{primitive variables}} \quad \& \quad \underbrace{\tilde{A} = \begin{pmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & \rho a^2 & u \end{pmatrix}}_{\text{Jacobain matrix of primitive variables}}$$

$$|\tilde{A} - \lambda I| = 0 \rightarrow \begin{vmatrix} u - \lambda & \rho & 0 \\ 0 & u - \lambda & \frac{1}{\rho} \\ 0 & \rho a^2 & u - \lambda \end{vmatrix} = 0 \rightarrow \begin{cases} \lambda_1 = \frac{dx}{dt} \Big|_1 = u \\ \lambda_1 = \frac{dx}{dt} \Big|_2 = u + a \\ \lambda_1 = \frac{dx}{dt} \Big|_3 = u - a \end{cases}$$

فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- تعیین فرم مشخصه‌ای سیستم معادلات

$$\tilde{l}_i \tilde{A}_{ij} = \tilde{l}_j \lambda \rightarrow (\tilde{l}_1, \tilde{l}_2, \tilde{l}_3) \begin{pmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & \rho a^2 & u \end{pmatrix} = \lambda (\tilde{l}_1, \tilde{l}_2, \tilde{l}_3)$$

$$\lambda_1 = u \rightarrow \begin{cases} u \tilde{l}_1 = u \tilde{l}_1 \\ \rho \tilde{l}_1 + u \tilde{l}_2 + \rho a^2 \tilde{l}_3 = u \tilde{l}_2 \\ \frac{1}{\rho} \tilde{l}_2 + u \tilde{l}_3 = u \tilde{l}_3 \end{cases} \rightarrow \begin{cases} \tilde{l}_1 = \alpha \\ \tilde{l}_2 = 0 \\ \tilde{l}_3 = -\frac{\alpha}{a^2} \end{cases}$$

$$\begin{cases} \lambda_1 = u \rightarrow (L^{(1)})^T = (\tilde{l}_1^{(1)}, \tilde{l}_2^{(1)}, \tilde{l}_3^{(1)}) = \left(\alpha, 0, -\frac{\alpha}{a^2} \right) \\ \lambda_2 = u + a \rightarrow (L^{(2)})^T = (\tilde{l}_1^{(2)}, \tilde{l}_2^{(2)}, \tilde{l}_3^{(2)}) = \left(0, \beta, \frac{\beta}{\rho a} \right) \\ \lambda_3 = u - a \rightarrow (L^{(3)})^T = (\tilde{l}_1^{(3)}, \tilde{l}_2^{(3)}, \tilde{l}_3^{(3)}) = \left(0, \delta, -\frac{\delta}{\rho a} \right) \end{cases} \rightarrow [L^{(i)}]^T = \begin{pmatrix} \tilde{l}_1^{(1)} & \tilde{l}_2^{(1)} & \tilde{l}_3^{(1)} \\ \tilde{l}_1^{(2)} & \tilde{l}_2^{(2)} & \tilde{l}_3^{(2)} \\ \tilde{l}_1^{(3)} & \tilde{l}_2^{(3)} & \tilde{l}_3^{(3)} \end{pmatrix} = \begin{pmatrix} 1 & 0 & -\frac{1}{a^2} \\ 0 & 1 & \frac{1}{\rho a} \\ 0 & 1 & -\frac{1}{\rho a} \end{pmatrix}$$

فرمولاسیون ریاضی معادلات اوایلر

- معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- تعیین متغیرهای مشخصه‌ای

$$\delta \hat{Q} = L^T \delta \tilde{Q} \quad \rightarrow \quad \begin{pmatrix} \delta \hat{Q}_1 \\ \delta \hat{Q}_2 \\ \delta \hat{Q}_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -\frac{1}{a^2} \\ 0 & 1 & \frac{1}{\rho a} \\ 0 & 1 & -\frac{1}{\rho a} \end{pmatrix} \begin{pmatrix} \delta \rho \\ \delta u \\ \delta p \end{pmatrix} \quad \rightarrow \quad \begin{cases} \delta \hat{Q}_1 = \delta \rho - \frac{1}{a^2} \delta p \\ \delta \hat{Q}_2 = \delta u + \frac{1}{\rho a} \delta p \\ \delta \hat{Q}_3 = \delta u - \frac{1}{\rho a} \delta p \end{cases}$$

$$\begin{cases} \hat{Q}_1 \text{ propagates with the speed } u \text{ along the characteristic } C: \frac{dx}{dt} = u \\ \hat{Q}_2 \text{ propagates with the speed } u + a \text{ along the characteristic } C^+: \frac{dx}{dt} = u + a \\ \hat{Q}_3 \text{ propagates with the speed } u - a \text{ along the characteristic } C^-: \frac{dx}{dt} = u - a \end{cases}$$

فرمولاسیون ریاضی معادلات اوایلر

$$\frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = 0$$

$$L^T \frac{\partial \tilde{Q}}{\partial t} + \underbrace{L^T \tilde{A} (L^T)^{-1}}_D L^T \frac{\partial \tilde{Q}}{\partial x} = 0$$

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- معادلات سازگاری و متغیرهای ریمان

$$\begin{pmatrix} 1 & 0 & -\frac{1}{a^2} \\ 0 & 1 & \frac{1}{\rho a} \\ 0 & 1 & -\frac{1}{\rho a} \end{pmatrix} \begin{pmatrix} \rho \\ u \\ p \end{pmatrix}_t + \begin{pmatrix} u & 0 & 0 \\ 0 & u+a & 0 \\ 0 & 0 & u-a \end{pmatrix} \begin{pmatrix} 1 & 0 & -\frac{1}{a^2} \\ 0 & 1 & \frac{1}{\rho a} \\ 0 & 1 & -\frac{1}{\rho a} \end{pmatrix} \begin{pmatrix} \rho \\ u \\ p \end{pmatrix}_x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} \frac{\partial \rho}{\partial t} - \frac{1}{a^2} \frac{\partial p}{\partial t} + u \left(\frac{\partial \rho}{\partial x} - \frac{1}{a^2} \frac{\partial p}{\partial x} \right) = 0 & \text{on } C: \frac{dx}{dt} = u \\ \frac{\partial u}{\partial t} + \frac{1}{\rho a} \frac{\partial p}{\partial t} + (u+a) \left(\frac{\partial u}{\partial x} + \frac{1}{\rho a} \frac{\partial p}{\partial x} \right) = 0 & \text{on } C^+: \frac{dx}{dt} = u+a \\ \frac{\partial u}{\partial t} - \frac{1}{\rho a} \frac{\partial p}{\partial t} + (u-a) \left(\frac{\partial u}{\partial x} - \frac{1}{\rho a} \frac{\partial p}{\partial x} \right) = 0 & \text{on } C^-: \frac{dx}{dt} = u-a \end{cases}$$

فرمولاسیون ریاضی معادلات اوایلر

On $C: \frac{dx}{dt} = u$

$$\frac{\partial \rho}{\partial t} - \frac{1}{a^2} \frac{\partial p}{\partial t} + u \left(\frac{\partial \rho}{\partial x} - \frac{1}{a^2} \frac{\partial p}{\partial x} \right) = 0$$

$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} - \frac{1}{a^2} \left(\frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} \right) = 0$$

$$\frac{d\rho}{dt} - \frac{1}{a^2} \frac{dp}{dt} = 0 \quad \text{or} \quad d\rho - \frac{1}{a^2} dp = 0$$

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- معادلات سازگاری و متغیرهای ریمان

$$s = k \ln \Omega \quad \& \quad k = \text{universal (Boltzmann) constant} = 1.38 \times 10^{-23} \text{ J/K}$$

$$s = c_v \ln e + R \ln \rho + \text{const.} \quad \& \quad s = c_v \ln p - c_p \ln \rho + \text{const.}$$

For homentropic: $s = \text{const.} \rightarrow s = c_v \ln p - c_p \ln \rho = \text{const.} \quad \text{or} \quad \frac{p}{\rho^\gamma} = \text{const.}$

$$ds = c_v \frac{dp}{p} - c_p \frac{d\rho}{\rho} = -\gamma \frac{c_v}{\rho} \left(d\rho - \frac{\rho}{\gamma p} dp \right) = -\gamma \frac{c_v}{\rho} \left(d\rho - \frac{1}{a^2} dp \right)$$

$$ds = -\gamma \frac{c_v}{\rho} \left(d\rho - \frac{1}{a^2} dp \right) = d\hat{Q}_1 = 0 \quad \text{on} \quad C: \frac{dx}{dt} = u \rightarrow \hat{Q}_1 = s$$

فرمولاسیون ریاضی معادلات اویلر

$$\text{On } C^+ : \frac{dx}{dt} = u + a$$

$$\frac{\partial u}{\partial t} + \frac{1}{\rho a} \frac{\partial p}{\partial t} + (u + a) \left(\frac{\partial u}{\partial x} + \frac{1}{\rho a} \frac{\partial p}{\partial x} \right) = 0$$

$$\frac{\partial u}{\partial t} + (u + a) \frac{\partial u}{\partial x} + \frac{1}{\rho a} \left(\frac{\partial p}{\partial t} + (u + a) \frac{\partial p}{\partial x} \right) = 0$$

$$\frac{du}{dt} + \frac{1}{\rho a} \frac{dp}{dt} = 0 \quad \text{or} \quad du + \frac{1}{\rho a} dp = 0 \quad \rightarrow \quad d\hat{Q}_2 = du + \frac{1}{\rho a} dp$$

$$\text{On } C^- : \frac{dx}{dt} = u - a$$

$$\frac{\partial u}{\partial t} - \frac{1}{\rho a} \frac{\partial p}{\partial t} + (u - a) \left(\frac{\partial u}{\partial x} - \frac{1}{\rho a} \frac{\partial p}{\partial x} \right) = 0$$

$$\frac{\partial u}{\partial t} + (u - a) \frac{\partial u}{\partial x} - \frac{1}{\rho a} \left(\frac{\partial p}{\partial t} + (u - a) \frac{\partial p}{\partial x} \right) = 0$$

$$\frac{du}{dt} - \frac{1}{\rho a} \frac{dp}{dt} = 0 \quad \text{or} \quad du - \frac{1}{\rho a} dp = 0 \quad \rightarrow \quad d\hat{Q}_3 = du - \frac{1}{\rho a} dp$$

• معادلات اویلر (فرم یک بعدی غیردائم تراکم پذیر)

- معادلات سازگاری و متغیرهای ریمان

فرمولاسیون ریاضی معادلات اوایلر

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- معادلات سازگاری و متغیرهای ریمان

$$\begin{cases} d\hat{Q}_2 = du + \frac{1}{\rho a} dp \\ d\hat{Q}_3 = du - \frac{1}{\rho a} dp \end{cases} \rightarrow \begin{cases} \hat{Q}_2 = u + \int \frac{1}{\rho a} dp & \text{on } C^+ : \frac{dx}{dt} = u + a \\ \hat{Q}_3 = u - \int \frac{1}{\rho a} dp & \text{on } C^- : \frac{dx}{dt} = u - a \end{cases}$$

For isentropic flow : $s = \text{const.} \rightarrow \frac{p}{\rho^\gamma} = \text{const.} = c \quad \& \quad dp = \frac{\partial p}{\partial \rho} d\rho = a^2 d\rho = \gamma c \rho^{\gamma-1} d\rho$

$$\begin{cases} \hat{Q}_2 = u + \int \frac{a}{\rho} d\rho = u + \int \frac{\sqrt{\gamma c}}{\rho} \rho^{\frac{\gamma-1}{2}} d\rho = u + \sqrt{\gamma c} \int \rho^{\frac{\gamma-3}{2}} d\rho = u + 2\sqrt{\gamma c} \frac{\rho^{\frac{\gamma-1}{2}}}{\gamma-1} = u + \frac{2}{\gamma-1} a & \text{on } C^+ \\ \hat{Q}_3 = u - \int \frac{a}{\rho} d\rho = u - \int \frac{\sqrt{\gamma c}}{\rho} \rho^{\frac{\gamma-1}{2}} d\rho = u - \sqrt{\gamma c} \int \rho^{\frac{\gamma-3}{2}} d\rho = u - 2\sqrt{\gamma c} \frac{\rho^{\frac{\gamma-1}{2}}}{\gamma-1} = u - \frac{2}{\gamma-1} a & \text{on } C^- \end{cases}$$

فرمولاسیون ریاضی معادلات اویلر

• معادلات اویلر (فرم یک بعدی غیردائم تراکم پذیر)

Compatibility eqs.:

- معادلات سازگاری و متغیرهای ریمان

$$\left\{ \begin{array}{l} \frac{d\hat{Q}_1}{dt} = \frac{\partial \hat{Q}_1}{\partial t} + u \frac{\partial \hat{Q}_1}{\partial x} = 0 \\ \frac{d\hat{Q}_2}{dt} = \frac{\partial \hat{Q}_2}{\partial t} + (u + a) \frac{\partial \hat{Q}_2}{\partial x} = 0 \\ \frac{d\hat{Q}_3}{dt} = \frac{\partial \hat{Q}_3}{\partial t} + (u - a) \frac{\partial \hat{Q}_3}{\partial x} = 0 \end{array} \right. \quad \text{on} \quad \begin{array}{l} C : \frac{dx}{dt} = u \\ C^+ : \frac{dx}{dt} = u + a \\ C^- : \frac{dx}{dt} = u - a \end{array}$$

Riemann variables:

$$\left\{ \begin{array}{l} \hat{Q}_1 = s \\ \hat{Q}_2 = u + \frac{2}{\gamma - 1} a \\ \hat{Q}_3 = u - \frac{2}{\gamma - 1} a \end{array} \right. \quad \text{on} \quad \begin{array}{l} C : \frac{dx}{dt} = u \\ C^+ : \frac{dx}{dt} = u + a \\ C^- : \frac{dx}{dt} = u - a \end{array}$$

فرمولاسیون ریاضی معادلات اوایلر

$$\frac{\partial \tilde{Q}}{\partial t} + \tilde{A} \frac{\partial \tilde{Q}}{\partial x} = 0$$

$$\tilde{Q} = \begin{pmatrix} \rho \\ u \\ p \end{pmatrix} \quad \& \quad \tilde{A} = \begin{pmatrix} u & \rho & 0 \\ 0 & u & \frac{1}{\rho} \\ 0 & \rho a^2 & u \end{pmatrix}$$

• معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

- رابطه مابین بردارهای چپ و راست ماتریس $\tilde{A}$

$$L^T = [L]^T = [L^{(1)}, L^{(2)}, L^{(3)}]^T = \begin{pmatrix} \tilde{l}_1^{(1)} & \tilde{l}_2^{(1)} & \tilde{l}_3^{(1)} \\ \tilde{l}_1^{(2)} & \tilde{l}_2^{(2)} & \tilde{l}_3^{(2)} \\ \tilde{l}_1^{(3)} & \tilde{l}_2^{(3)} & \tilde{l}_3^{(3)} \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1/a^2 \\ 0 & 1 & 1/(\rho a) \\ 0 & 1 & -1/(\rho a) \end{pmatrix}$$

$$\begin{cases} L^T \tilde{A} = D L^T & \& L^T \tilde{A} (L^T)^{-1} = D \\ \tilde{A} \tilde{R} = \tilde{R} D & \& (\tilde{R})^{-1} \tilde{A} \tilde{R} = D \end{cases} \rightarrow \tilde{R} = (L^T)^{-1} \quad \text{or} \quad \tilde{R} L^T = I$$

$$\tilde{R} = [\tilde{R}] = [\tilde{R}^{(1)}, \tilde{R}^{(2)}, \tilde{R}^{(3)}] = \begin{pmatrix} \tilde{r}_1^{(1)} & \tilde{r}_1^{(2)} & \tilde{r}_1^{(3)} \\ \tilde{r}_2^{(1)} & \tilde{r}_2^{(2)} & \tilde{r}_2^{(3)} \\ \tilde{r}_3^{(1)} & \tilde{r}_3^{(2)} & \tilde{r}_3^{(3)} \end{pmatrix} = (L^T)^{-1} = \begin{pmatrix} 1 & \rho/(2a) & -\rho/(2a) \\ 0 & 1/2 & 1/2 \\ 0 & \rho a/2 & -\rho a/2 \end{pmatrix}$$

فرمولاسیون ریاضی معادلات اوایلر

• روش جداسازی شار (Flux vector splitting method)

- معادلات اوایلر (فرم یک بعدی غیردائم تراکم پذیر)

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = 0$$

$$Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} \quad \& \quad F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix}$$

$$A = \frac{\partial F}{\partial Q} \quad \& \quad AR = RD \quad \& \quad A = R D R^{-1} = R(D^+ + D^-)R^{-1} = \underbrace{R D^+ R^{-1}}_{A^+} + \underbrace{R D^- R^{-1}}_{A^-}$$

$$F = A Q = (A^+ + A^-) Q = \underbrace{A^+ Q}_{F^+} + \underbrace{A^- Q}_{F^-}$$

$$\frac{\partial F}{\partial x} = \frac{\partial F^+}{\partial x} + \frac{\partial F^-}{\partial x} \rightarrow \frac{\partial F^+}{\partial x} = \begin{cases} \frac{F_i^+ - F_{i-1}^+}{\Delta x} + O(\Delta x) \\ \frac{3F_i^+ - 4F_{i-1}^+ + F_{i-2}^+}{2\Delta x} + O(\Delta x)^2 \end{cases} \quad \& \quad \frac{\partial F^-}{\partial x} = \begin{cases} \frac{F_{i+1}^- - F_i^-}{\Delta x} + O(\Delta x) \\ \frac{-F_{i+2}^- + 4F_{i+1}^- - 3F_i^-}{2\Delta x} + O(\Delta x)^2 \end{cases}$$

فرمولاسیون ریاضی معادلات اوایلر

• روش جداسازی شار (Flux vector splitting method)

- رابطه مابین $\tilde{R}$ و R $\rightarrow \tilde{R} = M^{-1} R$ $\tilde{A} = M^{-1} A M = M^{-1} (R D R^{-1}) M = \tilde{R} D (\tilde{R})^{-1}$

$$R = M \tilde{R} = M (L^T)^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ u & \rho & 0 \\ u^2/2 & \rho u & 1/(\gamma-1) \end{pmatrix} \begin{pmatrix} 1 & \rho/(2a) & -\rho/(2a) \\ 0 & 1/2 & 1/2 \\ 0 & \rho a/2 & -\rho a/2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \rho/(2a) & -\rho/(2a) \\ u & \rho(u+a)/(2a) & -\rho(u-a)/(2a) \\ u^2/2 & \rho(\chi+ua)/(2a) & -\rho(\chi-ua)/(2a) \end{pmatrix} \quad \& \quad \chi = u^2/2 + a^2/(\gamma-1)$$

$$R^{-1} = (\tilde{R})^{-1} M^{-1} = L^T M^{-1} = \begin{pmatrix} 1-u^2(\gamma-1)/(2a^2) & (\gamma-1)u/a^2 & -(\gamma-1)/a^2 \\ [(\gamma-1)u^2/2-ua]/(\rho a) & [a+(\gamma-1)u]/(\rho a) & (\gamma-1)/(\rho a) \\ [(\gamma-1)u^2/2+ua]/(\rho a) & -[a+(\gamma-1)u]/(\rho a) & (\gamma-1)/(\rho a) \end{pmatrix}$$

فرمولاسیون ریاضی معادلات اوایلر

• روش جداسازی شار (Flux vector splitting method)

- محاسبه شارهای F^+ و F^-

$$A = R D R^{-1} = R (D^+ + D^-) R^{-1} = \underbrace{R D^+ R^{-1}}_{A^+} + \underbrace{R D^- R^{-1}}_{A^-}$$

$$F = A Q = (A^+ + A^-) Q = \underbrace{A^+ Q}_{F^+} + \underbrace{A^- Q}_{F^-}$$

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = D^+ + D^-$$

For subsonic flow: $0 < u < a$ or $0 < M < 1$

$$D = \begin{pmatrix} u & 0 & 0 \\ 0 & u+a & 0 \\ 0 & 0 & u-a \end{pmatrix} = \underbrace{\begin{pmatrix} u & 0 & 0 \\ 0 & u+a & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{D^+} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & u-a \end{pmatrix}}_{D^-}$$

فرمولاسیون ریاضی معادلات اوایلر

• روش جداسازی شار (Flux vector splitting method)

- محاسبه شارهای F^+ و F^-

For subsonic flow: $0 < u < a$ or $0 < M < 1$

$$F^+ = A^+ Q = R D^+ R^{-1} Q = \frac{\rho}{2\gamma} \begin{pmatrix} 2\gamma u + a - u \\ 2(\gamma - 1)u^2 + (u + a)^2 \\ (\gamma - 1)u^3 + \frac{(u + a)^3}{2} + \frac{(3 - \gamma)(u + a)a^2}{2(\gamma - 1)} \end{pmatrix}$$

$$F^- = A^- Q = R D^- R^{-1} Q = \frac{\rho}{2\gamma} \begin{pmatrix} u - a \\ (u - a)^2 \\ \frac{(u - a)^3}{2} + \frac{(3 - \gamma)(u - a)a^2}{2(\gamma - 1)} \end{pmatrix} \quad \text{or} \quad F^- = F - F^+$$

فرمولاسیون ریاضی معادلات اوایلر

• روش جداسازی شار (Flux vector splitting method)

- محاسبه شارهای F^+ و F^-

For supersonic flow: $u > a$ or $M > 1$

$$D = \begin{pmatrix} u & 0 & 0 \\ 0 & u+a & 0 \\ 0 & 0 & u-a \end{pmatrix} = \underbrace{\begin{pmatrix} u & 0 & 0 \\ 0 & u+a & 0 \\ 0 & 0 & u-a \end{pmatrix}}_{D^+} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{D^-}$$

$$\begin{cases} A^+ = R D^+ R^{-1} = A \\ A^- = R D^- R^{-1} = 0 \end{cases} \rightarrow F^+ = A^+ Q = F = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix} \quad \& \quad F^- = A^- Q = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

فرمولاسیون ریاضی معادلات اوایلر

- روش جداسازی شار (Flux vector splitting method)

- محاسبه شارهای F^+ و F^-

For stagnation and sonic points: $M = 0$ ($u = 0$) & $M = 1$ ($u = a$)

$$\begin{cases} \lambda_k^+ = \frac{\lambda_k + \sqrt{\lambda_k^2 + \varepsilon^2}}{2} \\ \lambda_k^- = \frac{\lambda_k - \sqrt{\lambda_k^2 + \varepsilon^2}}{2} \end{cases} \quad \& \quad \varepsilon = \text{blending factor} \approx 0.01 - 0.04$$

فرمولاسیون ریاضی معادلات اوایلر

- معادلات اوایلر (فرم شبه خطی یک بعدی غیردائم تراکم پذیر، جریان درون نازل)

1D unsteady quasi-linear compressible Euler eqs. (nozzle flow):

$$\frac{\partial(SQ)}{\partial t} + \frac{\partial F}{\partial x} = H \quad \& \quad Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} \quad \& \quad F = S \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (E_t + p)u \end{pmatrix} \quad \& \quad H = \frac{dS}{dx} \begin{pmatrix} 0 \\ p \\ 0 \end{pmatrix} \quad \& \quad S = S(x)$$

$$\underbrace{\lambda_1 = u \quad \& \quad \lambda_2 = u + a \quad \& \quad \lambda_3 = u - a}_{\text{eigenvalues}}$$

$$\underbrace{\begin{cases} \frac{d\hat{Q}_1}{dt} = \frac{\partial \hat{Q}_1}{\partial t} + u \frac{\partial \hat{Q}_1}{\partial x} = 0 \\ \frac{d\hat{Q}_2}{dt} = \frac{\partial \hat{Q}_2}{\partial t} + (u + a) \frac{\partial \hat{Q}_2}{\partial x} = -\frac{ua}{S(x)} \frac{dS(x)}{dx} \\ \frac{d\hat{Q}_3}{dt} = \frac{\partial \hat{Q}_3}{\partial t} + (u - a) \frac{\partial \hat{Q}_3}{\partial x} = +\frac{ua}{S(x)} \frac{dS(x)}{dx} \end{cases}}_{\text{compatibility eqs.}}$$

$$\& \quad \underbrace{\begin{cases} \hat{Q}_1 = s & \text{on } C : \frac{dx}{dt} = u \\ \hat{Q}_2 = u + \frac{2}{\gamma-1}a & \text{on } C^+ : \frac{dx}{dt} = u + a \\ \hat{Q}_3 = u - \frac{2}{\gamma-1}a & \text{on } C^- : \frac{dx}{dt} = u - a \end{cases}}_{\text{Riemann variables}}$$

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (فرم شبه خطی یک بعدی غیردائم تراکم پذیر، جریان درون نازل)

Explicit flux-vector splitting scheme:

$$\frac{\partial(SQ)}{\partial t} + \frac{\partial F}{\partial x} = H$$

$$S_i \frac{Q_i^{n+1} - Q_i^n}{\Delta t} + \left(\frac{\partial F^+}{\partial x} + \frac{\partial F^-}{\partial x} \right)_i^n = H_i^n \quad \& \quad \Delta Q_i^n = Q_i^{n+1} - Q_i^n$$

$$S_i \frac{\Delta Q_i^n}{\Delta t} + \left(\frac{F_i^+ - F_{i-1}^+}{\Delta x} + \frac{F_{i+1}^- - F_i^-}{\Delta x} \right)^n = H_i^n$$

$$\Delta Q_i^n = -\frac{\Delta t}{S_i \Delta x} \left(F_i^+ - F_{i-1}^+ + F_{i+1}^- - F_i^- \right)^n + \frac{\Delta t}{S_i} H_i^n + O(\Delta t, \Delta x)$$

$$Q_i^{n+1} = Q_i^n + \Delta Q_i^n$$

فرمولاسیون ریاضی معادلات اوایلر

• روش جداسازی شار (Flux vector splitting method)

For subsonic flow: $0 \leq u \leq a$ or $0 \leq M \leq 1$

$$F^+ = A^+ Q = R D^+ R^{-1} Q = \frac{\rho}{2\gamma} S \begin{pmatrix} 2\gamma u + a - u \\ 2(\gamma - 1)u^2 + (u + a)^2 \\ (\gamma - 1)u^3 + \frac{(u + a)^3}{2} + \frac{(3 - \gamma)(u + a)a^2}{2(\gamma - 1)} \end{pmatrix}$$
$$F^- = A^- Q = R D^- R^{-1} Q = \frac{\rho}{2\gamma} S \begin{pmatrix} u - a \\ (u - a)^2 \\ \frac{(u - a)^3}{2} + \frac{(3 - \gamma)(u - a)a^2}{2(\gamma - 1)} \end{pmatrix} \quad \text{or} \quad F^- = F - F^+$$

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (فرم شبه خطی یک بعدی غیردائم تراکم پذیر، جریان درون نازل)

Implicit flux-vector splitting scheme:

$$\frac{\partial(SQ)}{\partial t} + \frac{\partial F}{\partial x} = H$$

$$S_i \frac{Q_i^{n+1} - Q_i^n}{\Delta t} + \left(\frac{\partial F^+}{\partial x} + \frac{\partial F^-}{\partial x} \right)_i^{n+1} = H_i^{n+1} \quad \& \quad \Delta Q_i^n = Q_i^{n+1} - Q_i^n$$

$$S_i \frac{\Delta Q_i^n}{\Delta t} + \left(\frac{F_i^+ - F_{i-1}^+}{\Delta x} + \frac{F_{i+1}^- - F_i^-}{\Delta x} \right)^{n+1} = H_i^{n+1}$$

$$S_i \Delta Q_i^n + \frac{\Delta t}{\Delta x} (F_i^+ - F_{i-1}^+ + F_{i+1}^- - F_i^-)^{n+1} = \Delta t H_i^{n+1} + O(\Delta t, \Delta x)$$

$$F^{n+1} = F^n + \left(\frac{\partial F}{\partial t} \right)^n \Delta t + O(\Delta t)^2 = F^n + \left(\frac{\partial F}{\partial Q} \right)^n \left(\frac{\partial Q}{\partial t} \right)^n \Delta t + O(\Delta t)^2 = F^n + A^n \left(\frac{\Delta Q^n}{\Delta t} + O(\Delta t) \right) \Delta t + O(\Delta t)^2$$

$$\begin{cases} F^{n+1} = F^n + A^n \Delta Q^n + O(\Delta t)^2 & \& A = \partial F / \partial Q \\ H^{n+1} = H^n + B^n \Delta Q^n + O(\Delta t)^2 & \& B = \partial G / \partial Q \end{cases}$$

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (فرم شبه خطی یک بعدی غیردائم تراکم پذیر، جریان درون نازل)

$$S_i \Delta Q_i^n + \frac{\Delta t}{\Delta x} (F_i^+ - F_{i-1}^+ + F_{i+1}^- - F_i^-)^{n+1} = \Delta t H_i^{n+1}$$

$$\begin{cases} (F_i^+)^{n+1} = (F_i^+)^n + (A_i^+)^n \Delta Q_i^n \\ (F_{i-1}^+)^{n+1} = (F_{i-1}^+)^n + (A_{i-1}^+)^n \Delta Q_{i-1}^n \end{cases} \quad \& \quad \begin{cases} (F_{i+1}^-)^{n+1} = (F_{i+1}^-)^n + (A_{i+1}^-)^n \Delta Q_{i+1}^n \\ (F_i^-)^{n+1} = (F_i^-)^n + (A_i^-)^n \Delta Q_i^n \end{cases} \quad \& \quad H_i^{n+1} = H_i^n + B_i^n \Delta Q_i^n$$

$$S_i \Delta Q_i^n + \frac{\Delta t}{\Delta x} \left\{ (A_i^+)^n \Delta Q_i^n - (A_{i-1}^+)^n \Delta Q_{i-1}^n + (A_{i+1}^-)^n \Delta Q_{i+1}^n - (A_i^-)^n \Delta Q_i^n \right\} - \Delta t B_i^n \Delta Q_i^n = RHS_i^n$$

$$RHS_i^n = -\frac{\Delta t}{\Delta x} (F_i^+ - F_{i-1}^+ + F_{i+1}^- - F_i^-)^n + \Delta t H_i^n$$

$$-\frac{\Delta t}{\Delta x} (A_{i-1}^+)^n \Delta Q_{i-1}^n + \left(S_i I + \frac{\Delta t}{\Delta x} (A_i^+ - A_i^-) - \Delta t B_i \right)^n \Delta Q_i^n + \frac{\Delta t}{\Delta x} (A_{i+1}^-)^n \Delta Q_{i+1}^n = RHS_i^n$$

فرمولاسیون ریاضی معادلات اویلر

- معادلات اویلر (فرم شبه خطی یک بعدی غیردائم تراکم پذیر، جریان درون نازل)

$$-\frac{\Delta t}{\Delta x} \left(A_{i-1}^+ \right)^n \Delta Q_{i-1}^n + \left(S_i I + \frac{\Delta t}{\Delta x} \left(A_i^+ - A_i^- \right) - \Delta t B_i \right)^n \Delta Q_i^n + \frac{\Delta t}{\Delta x} \left(A_{i+1}^- \right)^n \Delta Q_{i+1}^n = RHS_i^n$$

$$\bar{A}_i^n \Delta Q_{i-1}^n + \bar{B}_i^n \Delta Q_i^n + \bar{C}_i^n \Delta Q_{i+1}^n = \bar{D}_i^n$$

$$\bar{A}_i^n = -\frac{\Delta t}{2} A_{i-1}^n \Delta Q_{i-1}^n$$

$$\bar{B}_i^n = \left(S_i I + \frac{\Delta t}{\Delta x} \left(A_i^+ - A_i^- \right) - \Delta t B_i \right)^n$$

$$\bar{C}_i^n = \frac{\Delta t}{\Delta x} \left(A_{i+1}^- \right)^n$$

$$\bar{D}_i^n = RHS_i^n = -\frac{\Delta t}{\Delta x} \left(F_i^+ - F_{i-1}^+ + F_{i+1}^- - F_i^- \right)^n + \Delta t H_i^n$$

فرمولاسیون ریاضی معادلات اوایلر

- معادلات اوایلر (فرم شبه خطی یک بعدی غیردائم تراکم پذیر، جریان درون نازل)

$$\begin{bmatrix} [\bar{B}]_2 & [\bar{C}]_2 & & \\ [\bar{A}]_3 & [\bar{B}]_3 & [\bar{C}]_3 & \\ & [\bar{A}]_i & [\bar{B}]_i & [\bar{C}]_i \\ & & [\bar{A}]_{i \max - 2} & [\bar{B}]_{i \max - 2} & [\bar{C}]_{i \max - 2} \\ & & & [\bar{A}]_{i \max - 1} & [\bar{B}]_{i \max - 1} \end{bmatrix}^n \begin{bmatrix} \Delta Q_2 \\ \Delta Q_3 \\ \\ \Delta Q_{i \max - 2} \\ \Delta Q_{i \max - 1} \end{bmatrix}^n = \begin{bmatrix} [\bar{D}]_2 \\ [\bar{D}]_3 \\ \\ [\bar{D}]_{i \max - 2} \\ [\bar{D}]_{i \max - 1} \end{bmatrix}^n$$

$$[\bar{D}]_2 = [\bar{D}]_2 - [\bar{A}]_2 \Delta Q_1 \quad \& \quad [\bar{D}]_{i \max - 1} = [\bar{D}]_{i \max - 1} - [\bar{C}]_{i \max - 1} \Delta Q_{i \max}$$

موفق باشید.

$$Q = \begin{pmatrix} \rho \\ \rho u \\ E_t \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \\ Q_3 \end{pmatrix} = \begin{pmatrix} \tilde{Q}_1 \\ \tilde{Q}_1 \tilde{Q}_2 \\ \frac{\tilde{Q}_1^\gamma e^{\frac{\tilde{Q}_3}{c_v}}}{\gamma-1} + \frac{1}{2} \tilde{Q}_1 \tilde{Q}_2^2 \end{pmatrix} \quad \& \quad \tilde{Q} = \begin{pmatrix} \rho \\ u \\ s \end{pmatrix} = \begin{pmatrix} \tilde{Q}_1 \\ \tilde{Q}_2 \\ \tilde{Q}_3 \end{pmatrix}$$

$$s = c_v \ln p - c_p \ln \rho = c_v \left(\ln p - \frac{c_p}{c_v} \ln \rho \right) = c_v \ln \frac{p}{\rho^\gamma} \quad \rightarrow \quad e^{\frac{s}{c_v}} = \frac{p}{\rho^\gamma} \quad \rightarrow \quad p = \rho^\gamma e^{\frac{s}{c_v}}$$

$$E_t = \frac{p}{\gamma-1} + \frac{1}{2} \rho u^2 = \frac{p(\rho, s)}{\gamma-1} + \frac{1}{2} \tilde{Q}_1 \tilde{Q}_2^2 = \frac{\rho^\gamma e^{\frac{s}{c_v}}}{\gamma-1} + \frac{1}{2} \tilde{Q}_1 \tilde{Q}_2^2 = \frac{\tilde{Q}_1^\gamma e^{\frac{\tilde{Q}_3}{c_v}}}{\gamma-1} + \frac{1}{2} \tilde{Q}_1 \tilde{Q}_2^2$$